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USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23

23BS301D: DISCRETE MATHEMATICS AND GRAPH THEORY

(CSE/IT/AI&DS/AI&ML)

Time: 3 hours
70

Max. Marks:

PART-A Compulsory Questions (Each carry 2 Marks)

10 X 2 =20M

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|-----------|-----------|--|--------------|
| I. | a) | Define well-formed formula | (BL3) |
| | b) | Define Demorgan's law | (BL3) |
| | c) | Define equivalence relation. | (BL4) |
| | d) | Define cyclic permutation. | (BL4) |
| | e) | In how many different ways can a committee of 5 teachers and 4 students be selected from 9 teachers and 15 students? | (BL6) |
| | f) | Write Fibonacci relation and sequence. | (BL6) |
| | g) | Define order and size of a graph. | (BL3) |
| | h) | Define connected and disconnected graphs. | (BL3) |
| | i) | State Euler's formula. | (BL1) |
| | j) | Define complete Binary tree. | (BL1) |

PART-B (Answer either A or B from each Question)

5 X 10 =50M

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|--------------|-----------|---|-----------------|
| II.A. | 1) | Prove that for any three propositions p, q, r
$\{[p \vee (q \vee r)]\} \wedge \neg q \Rightarrow (p \vee r)$ | 5M (BL3) |
| | 2) | Find the PDNF and PCNF of the formula $p \vee [\neg p \rightarrow \{q \vee (\neg q \rightarrow r)\}]$. | 5M (BL3) |

OR

- | | | | |
|--------------|-----------|---|-----------------|
| II.B. | 1) | Show that the premises $a \rightarrow (b \rightarrow c)$, $d \rightarrow (b \wedge \neg c)$, $a \wedge d$ are inconsistent. | 5M (BL3) |
| | 2) | Test the validity of the following arguments | 5M (BL3) |

Sonia is watching TV .

If Sonia is watching TV, then she is not studying.

If she is not studying, then her father will not buy her a scooty.

Therefore, Sonia father will not buy a scooty

- | | | | |
|---------------|-----------|--|-----------------|
| III.A. | 1) | Let $A = \{1, 2, 3, 4, 6, 8, 12\}$ On A, define the partial ordering relation R by aRb if and only if $a b$. (i) Draw the Hasse diagram for ii) Write down the relation matrix for R. | 5M (BL4) |
| | 2) | Let $A = B = \{a, b, c, d\}$ and $R = \{(a,a), (a,c), (b,c), (c,a), (d,b), (d,d)\}$ and $S = \{(a,b), (b,c), (c,a), (c,b), (d,c)\}$ compute M_R and M_S also Draw the digraph and find RoS , SoR . | 5M (BL4) |

OR

III.B. 1) Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by aRb if and only if ' a ' is a multiple of ' b '. Represent the relation R as a matrix and draw its digraph. **5M (BL4)**

2) Define Lattice. Also write the properties of lattices. **5M (BL4)**

IV.A. 1) Find the number of Permutations of the letters of the word **MASSASAUGA**. In how many ways of these all A's are together? How many of them begin with S? **5M (BL6)**

2) How many committees of 5 members with a given chair person can be selected from the 12 persons. **5M (BL6)**

OR

IV.B. 1) Find the generating functions for the following sequence
i) $0, 1, -2, 3, -4, \dots$ ii) $0, 2, 6, 12, 20, 30, 42 \dots$ **5M (BL6)**

2) Solve the R.R $a_n - 7a_{n-1} + 10a_{n-2} = 4^n, n \geq 2, a_0 = 8, a_1 = 36$. **5M (BL6)**

V.A. 1) Explain adjacency matrix and incidence matrix for a graph with suitable example. **5M (BL3)**

2) If G is a simple graph in which every vertex has degree at least k , prove that G contains a path of length at least k **5M (BL3)**

OR

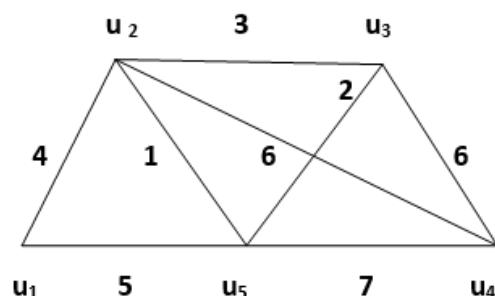
V.B. 1) Define isomorphism of two graphs. Explain with examples. **10M (BL3)**

VI.A. 1) Prove that C_6 is a bipartite graph **5M (BL1)**

2) How many vertices and how many edges are there in Complete Bipartite graphs $K_{4,7}$ and $K_{7,11}$ **5M (BL1)**

OR

VI.B. 1) Find the minimal spanning tree for the following graph using Prim's algorithm **10M (BL1)**



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(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23HS302: UNIVERSAL HUMAN VALUES – UNDERSTANDING
HARMONY AND ETHICAL HUMAN CONDUCT
(COMMON TO ALL BRANCHES)

Time: 3 hours
70

Max. Marks:

PART-A Compulsory Questions (Each carry 2 Marks)

10 X 2 =20M

- I. a)** Write three benefits of Value Education. **(BL2)**
- b)** Write 5 desires and categorize them into Right Understanding, Right Relationship and Physical Facility. **(BL5)**
- c)** What are the Responses of Self and Body? **(BL5)**
- d)** What is the program for nurturing and maintaining health of the body? **(BL3)**
- e)** Explain over evaluation, under evaluation and otherwise evaluation. **(BL5)**
- f)** What are the dimensions in the society? **(BL2)**
- g)** What is the inheritance of four orders in nature? **(BL2)**
- h)** Write about Existence. **(BL4)**
- i)** How do you implement value based living in individual level? **(BL3)**
- j)** Write two criteria for Technologies. **(BL2)**

PART-B (Answer either A or B from each Question)

5 X 10 =50M

- II.A. 1)** Make a list of your 20 desires. Now for each item on the list, find out Right Understanding, Right Relationship and Physical Facility. **10M (BL6)**

OR

- II.B. 1)** What is happiness? Give your own experience. **10M (BL3)**

III.A. 1) What are the needs of the self and the body? **10M (BL2)**

OR

III.B. 1) What are the programs for ensuring the health of the body? Explain. **10M (BL4)**

IV.A. 1) Give two examples each of over evaluation, under evaluation and otherwise evaluation. **10M (BL4)**

OR

IV.B. 1) What is the meaning of education and sanskāra? How does sanskāra follow education? **10M (BL5)**

V.A. 1) What do you mean by mutual fulfillment in nature? Cite a few examples. **10M (BL1)**

OR

V.B. 1) Suggest ways to enhance the fulfilment of human order with the other three orders. Mention any two programs you can undertake in light of the above. **10M (BL5)**

VI.A. 1) How does right understanding provide the basis for ethical human conduct? Give two examples. **10M (BL5)**

OR

VI .B. 1) How does right understanding provide the basis for humanistic constitution? Suggest some aspects of such a constitution to promote holistic living. **10M (BL3)**

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II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23ES303D: DIGITAL LOGIC & COMPUTER ORGANIZATION
(CSE/IT)

Time: 3 hours
70

Max. Marks:

PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M
I. a) Convert the $(7562.45)_8$ to Binary	(BL1)
b) What is a MUX. Draw a logic circuit of 8 X 1 multiplexer	(BL1)
c) How many FFs are needed for a mod 32 counter	(BL3)
d) Draw the logic symbol of negative edge triggered JK FF.	(BL5)
e) Represent +49 in 2's complement representation	(BL3)
f) Give advantages of hardwired control	(BL2)
g) Give the main difference between RAM and ROM	(BL2)
h) What is volatile and non-volatile memory	(BL2)
i) How I/O devices are interface to processor	(BL5)
j) List out the various I/O interface devices	(BL3)
 PART-B (Answer either A or B from each Question)	
5 X 10 =50M	
II.A. 1) What is a MUX. Draw a logic circuit of 8 X 1 multiplexer	10M (BL4)
OR	
II.B. 1) Using 2's complement perform subtraction (1101-1010)	5M (BL1)
2) What are the drawbacks of half adder circuit	5M (BL3)
III.A. 1) What are the differences between Sequential and Combinational logic circuits	10M (BL3)
OR	
III.B. 1) What is a counter and design a 4-bit Ripple up counter	10M (BL5)
IV.A. 1) Explain the addition and subtraction of signed magnitude numbers with examples with a flow chart.	10M (BL1)
OR	
IV.B. 1) How floating-point number are represented in computers	5M (BL3)
2) Explain how two floating-point are multiplied with an example	5M (BL2)
V.A. 1) A processor has 16 address lines and how much memory can it address and give its address map	10M (BL4)
OR	
V.B. 1) What is a virtual memory and explain the techniques of implementing it	10M (BL2)

VI.A. 1) Explain about IO BUS and Interface modules **10M (BL3)**

OR

VI 1) Compare and contrast various data transfer schemes used in **10M (BL2)**
.B. computer system

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II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23PCCS304: DATABASE MANAGEMENT SYSTEM
(CSE/AI&DS/AI&ML)

Time: 3 hours
70

Max. Marks:

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-
- PART-A Compulsory Questions (Each carry 2 Marks)** **10 X 2 =20M**
- I.**
- a)** Define the terms i) Entity set ii) Relationship set **(BL1)**
 - b)** Define data model. **(BL1)**
 - c)** What are the parts of SQL language? **(BL2)**
 - d)** List the set operations of SQL. **(BL2)**
 - e)** Write syntax on create command with example. **(BL2)**
 - f)** What is sub query **(BL2)**
 - g)** What meant by trivial dependency? **(BL1)**
 - h)** Define third normal form. **(BL1)**
 - i)** What are called secondary indices? **(BL2)**
 - j)** What is known as clustering file organization? **(BL2)**
- PART-B (Answer either A or B from each Question)** **5 X 10 =50M**
- II.A. 1)** Mention various groups of database users. Explain about their roles in Detail. **10M (BL4)**
- OR**
- II.B. 1)** Describe in detail about one-tier, two-tier and three-tier client-server architectures **10M (BL1)**
- III.A. 1)** Describe all operators of relational algebra. **10M (BL1)**
- OR**
- III.B. 1)** Write short notes on i) DDL ii) DML iii) DCL **10M (BL4)**
- IV.A. 1)** Illustrate Group by and Having clauses with examples? **10M (BL1)**
- OR**
- IV.B. 1)** What is SQL single row function? By means of suitable examples. Illustrate the usage of SQL date, character and number functions. **10M (BL3)**
- V.A. 1)** Explain briefly about 1NF, 2NF with suitable examples? **10M (BL4)**
- OR**
- V.B. 1)** Explain about Boyce Codd normal form (BCNF) with an example. **10M (BL4)**

VI.A. 1) Define transaction and explain desirable ACID properties of transactions **10M (BL1)**

OR

VI .B. 1) Describe about the view serializability and conflict serializability **10M (BL1)**

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II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23PCCS305: OBJECT ORIENTED PROGRAMMING THROUGH JAVA
(CSE/IT/AI&DS/AI&ML)

Time: 3 hours
70

Max. Marks:

PART-A Compulsory Questions (Each carry 2 Marks)

10 X 2 =20M

- I.**
- a) Java is Platform Independent language .Justify (BL5)
 - b) What is the difference between /(division) and % (modulus) operators . (BL2)
 - c) What is meant by private access specifier? (BL1)
 - d) What are the different types of constructors? (BL1)
 - e) What is a super class? (BL1)
 - f) Define abstract class? (BL1)
 - g) What is the difference between error and exception in java? (BL2)
 - h) What is ArrayIndexOutOfBoundsException? (BL2)
 - i) Give the difference between process and thread. (BL2)
 - j) Categorize any two methods to search a string in a String Class (BL4)

PART-B (Answer either A or B from each Question)

5 X 10 =50M

- II.A.**
- 1) Demonstrate Bitwise operator using a Java program. 5M (BL3)
 - 2) Develop a program to compute factorial of a number and explain the working 5M (BL6)

OR

- II.B.**
- 1) Explain in detail about jump statements in java 6M (BL2)
 - 2) Demonstrate logical operators with an example Java program 4M (BL3)

- III.A.**
- 1) Discuss about object oriented programming principles 10M (BL2)

OR

- III.B.**
- 1) Analyze about static variables and methods in detail. 10M (BL4)

- IV.A.**
- 1) Develop a Java program to perform Binary search. 10M (BL6)

OR

- IV.B.**
- 1) Write java program to demonstrate the single inheritance. 5M (BL6)
 - 2) Develop a Java program to implement multilevel inheritance 5M (BL6)

- V.A.**
- 1) Explain any three classes with method in Java.util package. 10M (BL1)

OR

- V.B.**
- 1) List and explain the try and catch blocks in Exception handling with examples. 5M (BL1)
 - 2) Explain why to use finally block in java exception handling 5M (BL1)

VI.A. 1) List and explain any five methods of a String class. **10M (BL1)**

OR

VI .B. 1) What is batch processing? What are the required methods in batch processing? **10M (BL1)**